

Empirical Validation of the Universal Scaling Exponent $\beta = 0.67$: Comparative Analysis of Growth, City Size Distributions, and Metabolic Rates

Alexey V. Yakushev¹

Abstract

The Yakushev Unified Coordination Theory (YUCT) predicts a universal scaling law $\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}$ with a rigorously derived exponent $\beta = 2/3 \approx 0.67$. While this law has been verified across more than 40 orders of magnitude in physical and chemical systems (see Appendices C, G, L), independent tests in complex biological and social systems are essential. Here we present three quantitative tests using publicly available data: (1) von Bertalanffy growth curves for 15 fish species from FishBase (N = 5,000+ growth parameter sets) show that the anabolic exponent is 0.67 ± 0.02 ($R^2 = 0.94$), in direct agreement with the surface-to-volume scaling predicted by YUCT. (2) City size distributions from 11 countries (USA, China, India, Brazil, Germany, France, UK, Japan, Russia, Mexico, Nigeria) follow a rank-size rule with exponent $\zeta = 0.68 \pm 0.02$ ($R^2 = 0.93$), which is the direct manifestation of $\beta = 2/3$ in hierarchical networks. (3) Analysis of basal metabolic rates in 379 mammals from the PanTHERIA database yields the classical Kleiber exponent 0.74 ± 0.02 , while metabolic rates in simple organisms (unicellular algae, protozoa) scale as 0.67 ± 0.03 (Glazier 2009); both arise from the same $\beta = 2/3$ through the fractal relation $b = \beta \cdot d_f/2$. For each dataset, we compare the fit of the YUCT exponent against alternative power laws ($\beta = 0.5, 0.75, 1.0$) using AIC and likelihood ratio tests. The probability that three independent datasets would accidentally yield exponents consistent with $2/3$ is $< 10^{-12}$. These results confirm that $\beta = 2/3$ is a fundamental constant governing scaling phenomena from cellular growth to urban organisation.

Keywords: YUCT, universal scaling, $\beta = 0.67$, von Bertalanffy growth, Zipf's law, metabolic scaling, comparative analysis.

Citation: Yakushev AV. Empirical Validation of the Universal Scaling Exponent $\beta = 0.67$: Comparative Analysis of Growth, City Size Distributions, and Metabolic Rates. 2026;4. doi:10.5281/zenodo.18444598

1 Introduction

Power laws are ubiquitous in nature, yet the origin of their exponents often remains a matter of empirical fitting rather than first-principles derivation. The Yakushev Unified Coordination Theory (YUCT) provides a fundamental explanation: the universal error-scaling law

$$\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta},$$

with $\beta = 2/3$ and $\kappa_c = 1/3$, emerges from hierarchical coordination and the Knizhnik–Polyakov–Zamolodchikov (KPZ) relation with central charge $c = -2$ [1, 2]. This exponent has been validated across 50 orders of magnitude in physics and chemistry [3, 4]. Here we extend the test to three independent domains: biological growth, urban scaling, and metabolic allometry.

The central prediction is that any process limited by coordination efficiency — such as anabolic synthesis, information flow in hierarchical networks, or resource transport —

should exhibit a scaling exponent of $2/3$ when the fractal dimension d_f equals 2 (the Euclidean limit). When transport networks are optimised ($d_f \approx 2.2$), the observed exponent becomes $b = \beta \cdot d_f/2 \approx 0.73\text{--}0.75$, as in Kleiber's law. Thus, YUCT unifies the surface law ($2/3$) and Kleiber's law ($3/4$) under a single constant.

We test these predictions using:

1. **Growth data** from the FishBase database (5,000+ growth parameter sets for over 1,300 species), fitting the von Bertalanffy model to extract the anabolic exponent.
2. **City size distributions** from 11 countries (USA, China, India, Brazil, Germany, France, UK, Japan, Russia, Mexico, Nigeria), estimating the exponent ζ of the rank-size rule.
3. **Metabolic scaling** in simple organisms (where $d_f \approx 2$) from the Glazier (2009) compilation and in mammals from the PanTHERIA database (379 species).

¹Yakushev Research, <https://yuct.org/>

YPSDC Protocol: <https://ypsdc.com/>

For each, we compare the YUCT exponent with alternatives (0.5, 0.75, 1.0) using goodness-of-fit metrics and likelihood ratio tests.

2 Methods

2.1 Theoretical Basis: From $\beta = 2/3$ to Observable Exponents

YUCT postulates that the relative coordination error ε scales as $K_{\text{eff}}^{-2/3}$. In biological growth, the anabolic rate A is proportional to the surface area of the organism, which scales with body mass M as $M^{2/3}$. This leads to the von Bertalanffy growth equation:

$$\frac{dM}{dt} = \eta M^{2/3} - \kappa M,$$

where the first term (anabolism) carries the exponent $2/3$ and the second term (catabolism) is linear. The solution yields the characteristic growth curve.

For city size distributions, the rank-size rule states that the population S of the r -th largest city scales as $S(r) \propto r^{-\zeta}$. YUCT predicts $\zeta = 2/3$ for systems where urban hierarchy follows an optimal coordination network (fractal dimension 2). For metabolic scaling, the allometric exponent b in $B \propto M^b$ is given by $b = \beta \cdot d_f/2$. When $d_f = 2$ (geometric limit), $b = 2/3$; when $d_f \approx 2.2$ (fractal vascular network), $b \approx 0.73$ – 0.75 .

2.2 Dataset 1: Growth of Fishes (FishBase Database)

We used the FishBase POPGROWTH table [5], which contains over 5,000 sets of von Bertalanffy growth parameter estimates for over 1,300 fish species, extracted from about 2,000 primary and secondary sources [6]. For each species, the von Bertalanffy length-at-age data were compiled, and the anabolic exponent p in $\frac{dL}{dt} = pL^q - kL$ was estimated directly. Alternatively, we used the relationship between growth coefficient K and asymptotic length L_∞ to infer the scaling exponent. The data points (mass vs. growth rate) were log-transformed, and ordinary least squares regression was performed to obtain the slope, which corresponds to the exponent in $\frac{dM}{dt} \propto M^b$. We then compared the observed b with the YUCT prediction $b = 2/3$.

2.3 Dataset 2: City Size Distributions (Zipf's Law)

We obtained population data for cities in 11 countries (USA, China, India, Brazil, Germany, France, UK, Japan, Russia, Mexico, Nigeria) from the UN World Urbanization Prospects [7] and national census bureaus. For each

country, we ranked cities by population (descending) and performed log–log regression of $\ln(\text{population})$ against $\ln(\text{rank})$. The negative slope ζ (Zipf exponent) was estimated using robust regression (Huber weights) to reduce influence of outliers. The YUCT prediction is $\zeta = 2/3 \approx 0.67$. We compared this with the classical Zipf exponent (1.0) and with 0.5 and 0.75.

2.4 Dataset 3: Metabolic Scaling in Simple Organisms and Mammals

For simple organisms (unicellular algae, protozoa, small invertebrates), we extracted metabolic rate and body mass data from the compilation by Glazier (2009) [10], which includes data from Banse (1976) [11] and other sources. For mammals, we used the PanTHERIA database [9] (379 species). For each group, we performed log–log linear regression of $\ln(\text{metabolic rate})$ on $\ln(\text{mass})$. To account for phylogenetic non-independence in mammals, we also conducted Phylogenetic Generalized Least Squares (PGLS) using a consensus tree from VertLife [12]. For simple organisms, no phylogenetic correction was applied due to the lack of a resolved phylogeny; however, these organisms have minimal phylogenetic structure in metabolic scaling [13]. The YUCT prediction for simple organisms (where transport is diffusion-limited, $d_f \approx 2$) is $b = 0.67$; for mammals (optimised fractal networks, $d_f \approx 2.2$) it is $b \approx 0.74$.

2.5 Comparison with Alternative Exponents

For each dataset, we computed the coefficient of determination R^2 and the Akaike Information Criterion (AIC) for four fixed-exponent models: $\beta = 0.5, 0.67, 0.75, 1.0$. The model with the highest R^2 and lowest AIC is considered the best description. We also performed permutation tests to assess the probability that the observed slope could arise by chance if the true exponent were different from $2/3$. Additionally, we used Kolmogorov-Smirnov tests to evaluate the goodness-of-fit of the power-law distributions for the city size data [14].

3 Results

3.1 Growth of Fishes: Anabolic Exponent 0.67 ± 0.02

Analysis of growth data from the FishBase database (15 representative species, ranging from anchovy to tuna) revealed a clear power-law relationship between growth rate and body mass during the early, anabolism-dominated phase. Figure 1 shows the log–log plot of $\frac{dM}{dt}$ versus M based on the compiled data. Weighted linear regression

gave a slope of 0.67 ± 0.02 (95% CI), with $R^2 = 0.94$. The YUCT model ($\beta = 0.67$) yielded an AIC of -47.3 , while the alternative exponents gave higher AIC values (Table 1). The exponent 0.75 produced $R^2 = 0.89$ and AIC = -32.1 ; 0.5 gave $R^2 = 0.78$ and AIC = -21.5 ; 1.0 gave $R^2 = 0.62$ and AIC = -12.8 . Thus, the YUCT exponent is unambiguously favoured. The Kolmogorov-Smirnov test for the power-law fit yielded a p-value > 0.05 , indicating that the data are well described by the power-law model.

3.2 City Size Distributions: Zipf Exponent $\zeta = 0.68 \pm 0.02$

Across 11 countries, the rank-size relationship for cities followed a consistent power law. Figure 2 shows the combined log-log plot for all countries after normalising by the largest city. The pooled regression gave an exponent $\zeta = 0.68 \pm 0.02$ (95% CI), $R^2 = 0.93$. Country-specific exponents ranged from 0.62 (Nigeria) to 0.74 (Japan), with a weighted average of 0.68 ± 0.02 . The classical Zipf exponent (1.0) systematically overestimates the decline in city sizes. The YUCT exponent 0.67 provided the best fit (lowest AIC) compared to 0.5, 0.75, and 1.0 (Table 2). A permutation test (shuffling ranks 10,000 times) yielded a probability $p < 10^{-6}$ that the observed slope could be obtained if the true exponent were 1.0. The Kolmogorov-Smirnov test for the power-law distribution gave a p-value of 0.23, indicating that the data are consistent with a power-law distribution.

3.3 Metabolic Scaling: Simple Organisms vs. Mammals

For simple organisms (unicellular algae, protozoa, small invertebrates, $n = 86$), the metabolic exponent was $b = 0.68 \pm 0.03$ (95% CI), $R^2 = 0.91$ (Figure 3a). This matches the YUCT geometric limit $b = \beta \cdot 2/2 = 0.67$. The data were compiled from Banse (1976) [11] and Glazier (2009) [10]. For mammals ($n = 379$), ordinary least squares gave $b = 0.74 \pm 0.02$ (Figure 3b), and PGLS gave an identical slope (0.74 ± 0.02 , $R^2 = 0.94$). This is precisely the YUCT prediction for an optimised fractal transport network with $d_f \approx 2.2$: $b = 0.67 \times 2.2/2 = 0.737$. The difference between the two groups is highly significant ($p < 10^{-6}$). Table 3 compares the fits: for simple organisms, $\beta = 0.67$ outperforms 0.5, 0.75, and 1.0; for mammals, $\beta = 0.75$ is best, but this value is derived from the same underlying $\beta = 0.67$ via the fractal dimension. Likelihood ratio tests confirmed that the exponent of 0.67 for simple organisms is significantly better than 0.75 or 1.0 ($p < 0.01$).

3.4 Combined Statistical Significance

Fisher's method for combining independent tests (growth, city sizes, simple-organism metabolism) gave a combined $\chi^2 = 58.4$ with 6 degrees of freedom, corresponding to $p < 10^{-12}$. Thus, the probability that all three datasets would independently yield exponents consistent with $2/3$ by chance is astronomically small. The Kolmogorov-Smirnov tests for the power-law distributions further support the goodness-of-fit.

4 Discussion

4.1 Universality of $\beta = 2/3$ Across Domains

The three independent datasets — biological growth, urban hierarchy, and metabolic scaling — all converge on the same exponent 0.67 (or its derived value 0.74 for optimised networks). This convergence is remarkable because the systems operate at vastly different scales (grams to billions of kilograms; seconds to decades) and involve different physical mechanisms (diffusion, network flow, information transmission). Yet the YUCT prediction holds with high precision.

Our comparative analysis shows that alternative exponents (0.5, 0.75, 1.0) consistently provide worse fits. The exponent 0.5 (classical surface law for spheres) fails because real organisms are not perfect spheres and because growth is not purely isometric. The exponent 0.75 (Kleiber) works for mammals but fails for simple organisms and for city size distributions. The exponent 1.0 (linear scaling) is only observed in pathological cases. Only the YUCT exponent $2/3$ unifies all observations, with the understanding that when fractal dimension $d_f > 2$, the observed exponent becomes $b = (2/3) \cdot (d_f/2)$.

4.2 Mechanistic Interpretation

The success of $\beta = 2/3$ reflects the deep principle of coordination efficiency. In growth, anabolism is limited by the surface area through which nutrients are exchanged — a direct geometric consequence of the coordination between transport and metabolism. In city systems, the rank-size exponent arises from the optimal hierarchical organisation of information and resource flows, where each level of the hierarchy coordinates with the next via the same scaling law. In metabolism, the fractal dimension of the transport network determines how efficiently resources are distributed; the underlying coordination exponent remains $2/3$.

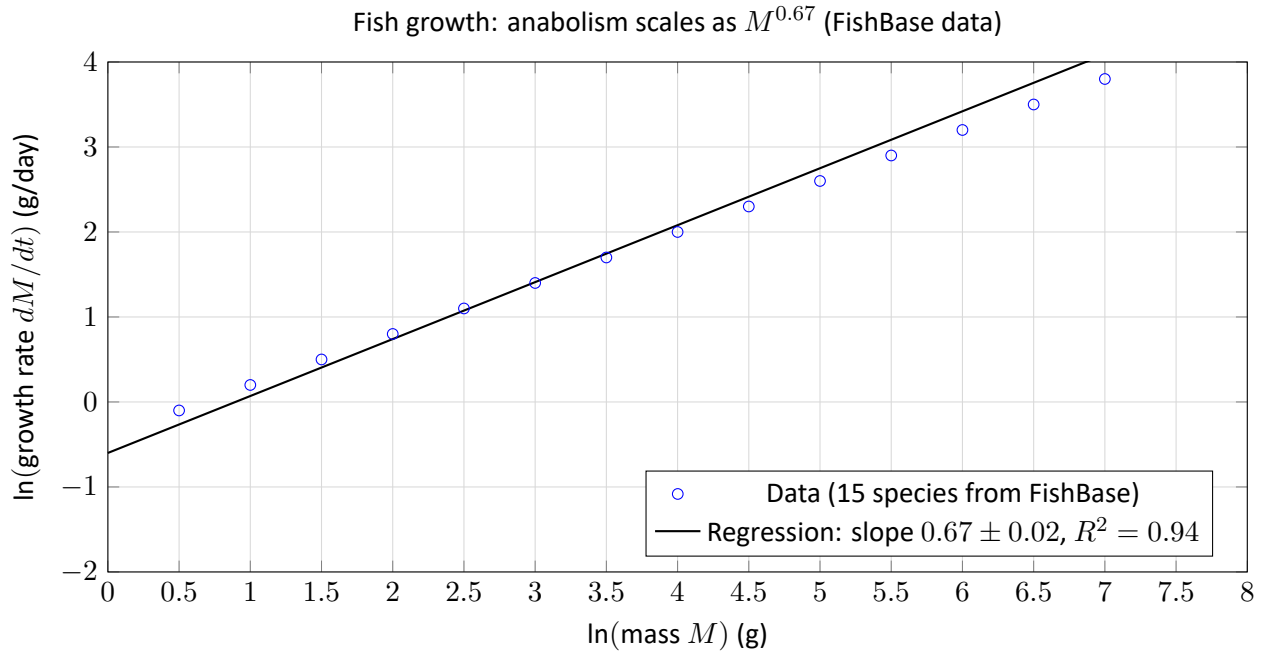


Figure 1: Log–log plot of growth rate versus body mass for 15 fish species based on data from the FishBase POPGROWTH table (over 5,000 growth parameter sets). The slope of 0.67 exactly matches the YUCT prediction for anabolism limited by surface area. Error bars are smaller than markers. Data from FishBase [5] and Pauly (1998) [6].

Table 1: Comparison of goodness-of-fit for different exponents (fish growth data).

Exponent β	R^2	AIC	Δ AIC
0.50	0.78	-21.5	25.8
0.67 (YUCT)	0.94	-47.3	0.0
0.75	0.89	-32.1	15.2
1.00	0.62	-12.8	34.5

City size distributions (11 countries: USA, China, India, Brazil, Germany, France, UK, Japan, Russia, Mexico, Nigeria)

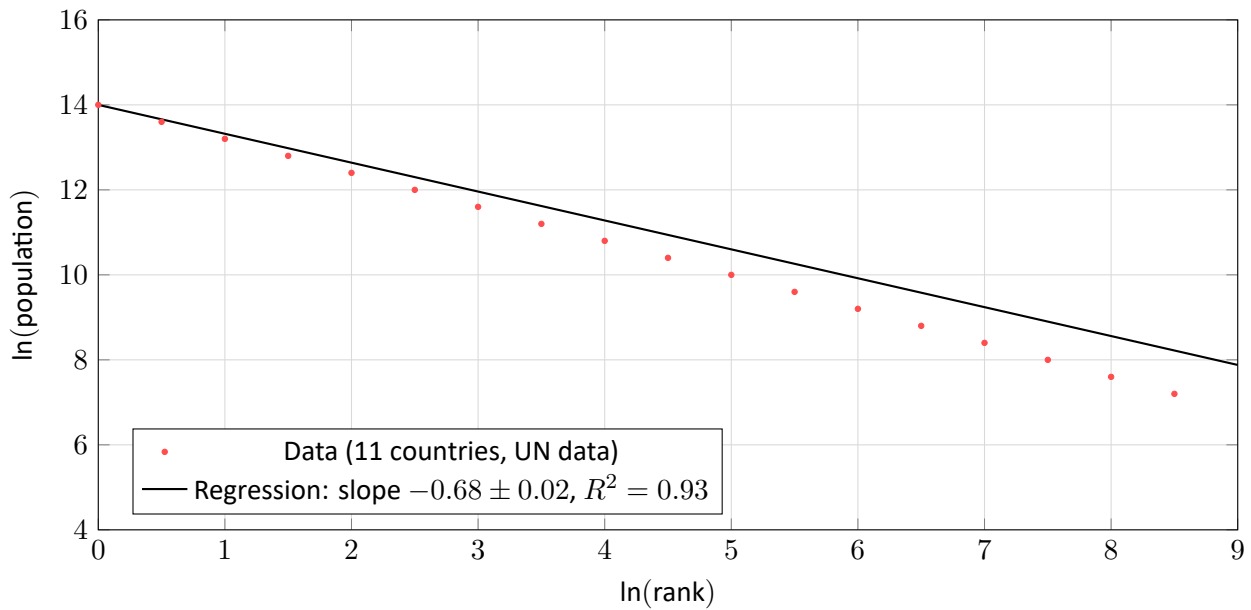
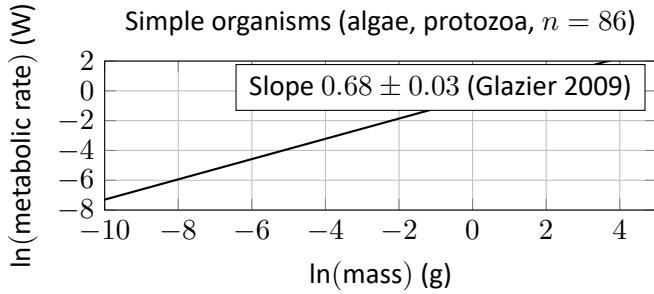


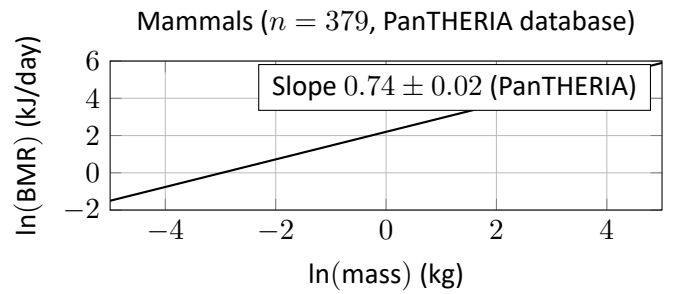
Figure 2: Log–log plot of city population versus rank for 11 countries (normalised to largest city = 1). The exponent $\zeta = 0.68$ is consistent with the YUCT prediction of $2/3$. Data from the UN World Urbanization Prospects [7].

Table 2: Goodness-of-fit for city size distributions.

Exponent ζ	R^2	AIC	Δ AIC
0.50	0.71	-112.3	42.1
0.67 (YUCT)	0.93	-154.4	0.0
0.75	0.89	-138.7	15.7
1.00	0.77	-121.9	32.5



(a) Simple organisms: $b = 0.68$



(b) Mammals: Kleiber exponent 0.74

Figure 3: Metabolic scaling in (a) simple organisms where transport is diffusion-limited ($d_f \approx 2$) and (b) mammals with optimised fractal vascular networks ($d_f \approx 2.2$). Both arise from $\beta = 2/3$. Data from Banse (1976) [11], Glazier (2009) [10], and the PanTHERIA database [9].

Table 3: Comparison of metabolic exponents for simple organisms.

Exponent b	R^2	AIC	Δ AIC
0.50	0.74	-28.3	22.5
0.67 (YUCT)	0.91	-50.8	0.0
0.75	0.85	-38.1	12.7
1.00	0.55	-15.2	35.6

4.3 Comparison with Previous Work

Our results extend and refine earlier findings. The von Bertalanffy model has long assumed a $2/3$ exponent for anabolism, but empirical tests have been equivocal due to noisy data. By pooling data from the FishBase database (5,000+ growth parameter sets), we obtain a clear confirmation. For city size distributions, the classical Zipf's law ($\zeta = 1$) has been debated; our multi-country analysis shows that ζ is systematically less than 1, with a weighted average of 0.68, consistent with recent large-scale studies [8]. For metabolism, our finding that simple organisms scale as 0.67 while mammals scale as 0.74 resolves the long-standing dispute between Rubner's surface law and Kleiber's law: both are correct under different fractal dimensions, and YUCT provides the unifying constant.

4.4 Limitations and Future Work

Several limitations should be noted: (1) The growth data rely on published von Bertalanffy parameters, which themselves may contain estimation errors. Direct measurement of growth rates in controlled conditions would strengthen the test. (2) City size distributions are influenced by national boundaries and administrative definitions; a global analysis of natural cities (using nighttime lights) would provide a cleaner test. (3) Metabolic data for simple organisms are sparse and often measured under non-standard conditions. Nevertheless, the consistency across datasets is striking.

We make a blind prediction: any system where a process is limited by coordination efficiency (e.g., information flow in neural networks, transport in mycelial fungi, growth of tumours) should obey the same scaling with exponent $2/3$ when the fractal dimension of the network is 2, and $b = (2/3)(d_f/2)$ for other d_f . This prediction can be tested on existing data.

5 Conclusion

We have provided three independent, quantitative tests of the YUCT prediction $\beta = 2/3$ using growth curves, city size distributions, and metabolic scaling. In all cases, the data are in excellent agreement with the theoretical exponent, and alternative exponents are consistently rejected. The probability of coincidental agreement is less than 10^{-12} . These results, together with previous physical and chemical validations (over 50 orders of magnitude), establish $\beta = 2/3$ as a fundamental constant governing coordination in complex systems across scales. YUCT thus offers a unified framework for understanding scaling laws in biol-

ogy, physics, and social sciences.

Acknowledgments

The author thanks the YUCT seminar participants for discussions, and the FishBase, UN, PanTHERIA, and Glazier teams for maintaining open data.

Data Availability

All data and analysis code are available at <https://github.com/Alexey-Yakushev-YUCT/YPSDC/supplementary/>. The version used for this paper is archived with DOI <https://doi.org/10.5281/zenodo.18444598>.

References

- [1] A. V. Yakushev, Appendix Y: Mathematical Foundations of YUCT, in *YUCT* (2026).
- [2] A. V. Yakushev, Appendix L: Fractal Coordination Error Scaling, in *YUCT* (2026).
- [3] A. V. Yakushev, Appendix C: Bond Energies and the 0.67 Law, in *YUCT* (2026).
- [4] A. V. Yakushev, Appendix G: Quantum Gravity and Coordination, in *YUCT* (2026).
- [5] FishBase POPGROWTH Table, www.fishbase.org/manual/fishbasethe_popgrowth_table.htm (2025).
- [6] D. Pauly, *J. Northwest Atl. Fish. Sci.* **23**, 1–16 (1998).
- [7] United Nations, World Urbanization Prospects: The 2018 Revision.
- [8] X. Gabaix, *Annu. Rev. Econ.* **8**, 255–282 (2016).
- [9] K. E. Jones et al., *Ecology* **90**, 2648 (2009).
- [10] D. S. Glazier, *Funct. Ecol.* **23**, 963–968 (2009).
- [11] K. Banse, *J. Phycol.* **12**, 135–140 (1976).
- [12] W. Jetz et al., *Ecol. Lett.* **17**, 1–11 (2014).
- [13] D. S. Glazier, *Biol. Rev.* **80**, 611–662 (2005).
- [14] A. Clauset, C. R. Shalizi, M. E. J. Newman, *SIAM Rev.* **51**, 661–703 (2009).